

CERN highlights in 2024 celebrating 70 years

For its 70th anniversary year, CERN honoured its contributions to fundamental research, innovation and international collaboration



A breathtaking sunrise illuminating the iconic CERN site. (Image: M.Brice/CERN)

2024 held special significance for CERN, as it celebrated its [70th anniversary](#). To mark this major milestone, the Organization held a series of public events in the CERN Science Gateway around the overarching themes: Unveiling the Universe, Exploring Farther: Machines for New Knowledge, Virtuous Circle of Knowledge and Innovation and Extraordinary Human Endeavour. It also held a [CERN70 Community Event](#) and an [Official Ceremony](#) for Heads of State and Government, along with [celebrations organised in the Member and Associate Member States](#) across the globe. The Office for Alumni Relations hosted its [Third Collisions event](#), a major reunion that welcomed 600 alumni at CERN.

Last year, CERN experiments have continued **Unveiling the Universe** by pushing the boundaries of our understanding. The [ATLAS and CMS experiments observed quantum entanglement at the highest energy yet](#) at the Large Hadron Collider (LHC), opening up a new perspective on the complex world of quantum physics. The ATLAS experiment harnessed machine learning to gain [new insights into Higgs self-interaction](#), and the CMS experiment achieved the [most precise measurement of the W boson mass](#) to date.

A Word from Fabiola Gianotti

Happy 2025 to the CERN community!

2025 will be a crucial year for CERN, with the work towards the update of the European Strategy for Particle Physics, the last full year of Run 3 and preparations for the HL-LHC era

Contents

News.....p.1–9

CERN highlights in 2024 celebrating 70 years

Accelerator Report: Another exciting year ahead

The measurement of a lifetime

LHCb sheds light on two pieces of the matter–antimatter puzzle

Work Well Feel Well: Emotions: forces for transformation

Computer Security: Are you a team player?

Official Communications.....p.9–12

Remuneration payment dates in 2025

Annual adjustments to financial benefits with effect from 1 January 2025

CERN Health Insurance Scheme (CHIS) --

Monthly contributions as of 1 January 2025

Staff Rules and Regulations - 11th edition

Composition of the Joint Advisory Appeals Board (JAAB/CPCR)

Revised computer security rules

Announcements.....p.12–16

Upcoming events

Find out who won the 2024 End-of-Year Crossword

Sign up for the 2025 Mentoring@CERN programme

Using portable heaters at CERN

Electrical Safety Awareness Day

Works on the Route du Nant-d'Avril from 13 January 2025

Sign up for CERN's Environmentally Responsible Procurement e-learning course

A word from Fabiola Gianotti

Happy 2025 to the CERN community!

2025 will be a crucial year for CERN, with the work towards the update of the European Strategy for Particle Physics, the last full year of Run 3 and preparations for the HL-LHC era

Happy 2025 to you all! I wish you, as well as your families, a year filled with health, happiness and success. May this year also bring well-being and personal fulfilment and growth to each of you.

In 2025, we have exciting new challenges to tackle and many opportunities to seize. And I am confident that, together, we will continue to make CERN and our field shine.

In the [New Year presentation on 14 January](#), the Directors and I gave a few examples of the great accomplishments in 2024. The year saw the extraordinary performance of the accelerator complex, the experiments and the computing. It was the best production year for proton–proton physics at the LHC ever, and the experiments collected about 220 PB of raw data, compared to about 100 PB in 2023. Many beautiful physics results were produced across the full scientific programme.

CERN's 70th anniversary was celebrated with a variety of events at CERN and in 33 countries, including an event for the CERN community on 17 September and the ceremony with political authorities on 1 October. CERN Science Gateway, in its first year, attracted more than 388 000 visitors from 175 countries and was featured as [THE place to visit](#) by media worldwide.

Looking to the year ahead, the top five objectives set by the Enlarged Directorate are as follows:

1. **Comply with the new schedule for the HL-LHC:** The [accelerator schedule](#) was updated in 2024, with the start of the HL-LHC pushed back by one year. Our first priority for 2025 is to maintain this schedule. Work on the accelerator and experiment upgrades and preparation for the immense amount of work to be carried out in Long Shutdown 3 (LS3) should be the major focus.

2. **Successful and safe operation of accelerators, experiments and computing:** Preliminary luminosity targets for 2025 have been set and will be finalised at the [Chamonix Workshop 2025](#) in late January.
3. **Complete the FCC Feasibility Study:** Great progress has been made across all aspects of the Study, and the final report should be completed by 31 March 2025.
4. **European Strategy for Particle Physics:** The [update of the European Strategy for Particle Physics](#) will be a major effort, and strong participation, notably from our young colleagues, is needed to ensure a brilliant future for CERN.
5. **Finalise the implementation of [CERN's main objectives for 2021–2025](#):** these include, in addition to scientific goals, aspects such as industrial and human resources return to Member and Associate Member States, personnel well-being, training, education and outreach, environment and sustainability and CERN's impact on society.

During 2025, CERN's organisational structure and Management for the period 2026–2030 will be established by the Director-General-elect, Mark Thomson, and I look forward to a smooth transition at the end of 2025.

2024 was a spectacular year for the Laboratory and its worldwide community and I would like to thank each one of you for your hard work, dedication and commitment to CERN. You have accomplished phenomenal achievements, and I look forward to another year of great success with all of you. I wish you and your loved ones a fantastic 2025.

Fabiola Gianotti
CERN Director-General

>>> *Continued from page 1*

The [ALICE experiment conducted the famous double-slit experiment](#) and found the [first ever evidence of the heaviest antimatter hypernucleus](#) at the LHC. The LHCb experiment [released its full data from LHC Run 1](#), thus giving open access to a wide range of physics studies, and probed [new searches for matter–antimatter asymmetry](#). The BASE experiment [successfully transported a box filled with unbonded protons](#) across CERN’s main site, thus demonstrating that the same feat could later be possible for antiprotons. The AegIS experiment paved the way for a new set of antimatter studies by [cooling positronium with laser light for the first time](#). Other experiments, such as [AMBER](#) explored dark matter searches, [SHiP set sail to explore the hidden sector](#) and [NA62 observed an ultra-rare particle decay](#). [CLOUD resolved mysteries about aerosol particles](#) emitted by tropical rainforests, and [MoEDAL set new limits on the mass of magnetic monopole](#), considerably narrowing the search window for these hypothetical particles.

In 2024, the LHC more than fulfilled its role of **Exploring Farther: Machines for New Knowledge** by having its most efficient year of operation to date, with [record physics data delivered](#). The experiments are upgrading their detectors in preparation for the High-Luminosity LHC (HL-LHC), where the project teams successfully completed the [installation of inner-triplet test string magnets](#) and [tests of the cold powering system](#).

CERN’s innovations have reverberated far beyond particle physics exemplifying the **Virtuous Circle of Knowledge and Innovation**. CERN brought together [key stakeholders in global health](#) and one of the flagship projects known as STELLA is [re-engineering radiotherapy](#) to make it accessible for low- and middle-income countries. A collaboration with EUROfusion is driving forward innovative

technologies for [future colliders and nuclear fusion reactors](#). AI-driven projects like [Edge SpAlce will help tackle marine plastic pollution](#), leveraging CERN’s expertise in data management, and the White Rabbit Collaboration has launched a new open-source technology to sync devices in the accelerators down to sub-nanoseconds, promising applications far beyond particle physics. The Organization also celebrated the first anniversary of [CERN Venture Connect](#) initiative, a launchpad for deep-tech startups, and the [tenth anniversary of the CERN and Society Foundation](#).

[CERN inaugurated a new data centre](#) with integrated heat-recovery system to address the data-processing needs of the worldwide scientific community and launched the [Next-Generation Triggers project](#) to significantly increase the efficiency, sensitivity and modelling of the ATLAS and CMS experiments.

CERN has reaffirmed its commitment to environmentally responsible research with a strategic partnership with ABB that [helps identify an energy-saving potential of 17.4% across cooling and ventilation motors](#) and [introduces a novel approach using carbon dioxide cooling technology](#) in order to minimise the Laboratory’s carbon footprint.

CERN’s **Extraordinary Human Endeavour** expanded as [Estonia became the 24th Member State](#). The [CERN Council appointed Professor Mark Thomson as the next Director-General](#), starting in 2026, and [Professor Costas Fountas as the next President](#).

Having marked seven decades of discovery in 2024, CERN now looks forward to another 70 extraordinary years of scientific research and global collaboration and welcomes 2025, the [International Year of Quantum Science and Technology](#).

Chetna Krishna

Accelerator Report: Another exciting year ahead

Although the much-appreciated winter break is well behind us and most of us are back at work, the accelerator complex is still in hibernation – but not for much longer

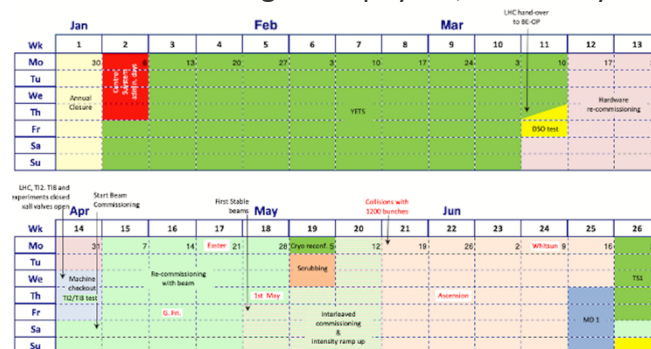
Just as the end of a run gives us the chance to look back and reflect on our achievements, the start of a new year is the perfect opportunity to focus on the plans and challenges ahead. The maintenance activities scheduled for the 2024–2025 year-end technical stop (YETS) are in full swing and, so far, no unexpected issues or showstoppers that could delay the restart of the accelerator complex have arisen.

Every year, the accelerator complex is restarted through a carefully planned process that combines sequential and parallel workflows. This approach optimises the time available for YETS maintenance activities, while maximising physics time during the operational run. The goal for the recommissioning is to deliver all beams on schedule, with the quality and performance required by the experiments.

For the LHC, maintenance activities are scheduled to conclude on 13 March, when responsibility for the LHC will transfer from the Technical Coordination group in the Engineering department (EN-ACE) to the Operations group in the Beams department (BE-OP). The first crucial milestone in preparing the LHC for operation will be the personnel access and safety system tests, conducted by the Beams Departmental Safety Officer and her team (DSO tests). These tests will enable us to verify and validate the functioning of the safety system, which is designed to ensure that no one can be inside the LHC when it is operational and that the machine cannot be started if personnel are present in the tunnel.

Following the validation of these safety systems, the LHC's hardware will be powered up and thoroughly tested over a 2.5-week period. This process will culminate in a three-day machine check-out, scheduled to begin on 2 April. During this period, the LHC will be operated as if beams were circulating, but without the actual beam, a bit like an orchestra's dress rehearsal, where all the instruments play together even if they are not yet perfectly tuned. Once the "orchestra" has been fine-tuned, the LHC will be ready to receive its "musical score", the proton beam, and begin its symphony of accelerating and colliding beams.

Before the full physics programme begins, a six-week period of beam recommissioning and intensity ramp-up will take place, bringing us to 1200 bunches per beam, a number that is considered meaningful for physics, on 19 May.

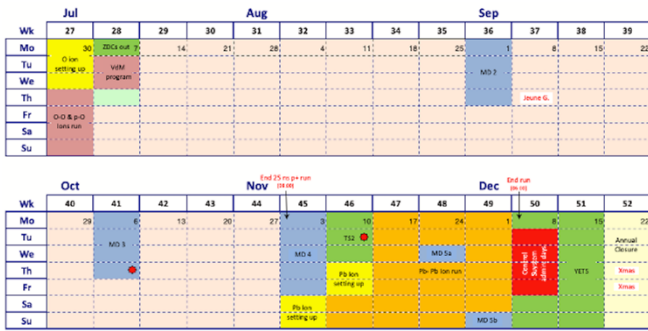


The first two quarters of the approved 2025 LHC schedule. The green part until the end of week 11 represents the YETS, followed by the hardware recommissioning and machine check-out in weeks 12, 13 and 14. First beam injection is scheduled for 5 April, followed by the beam recommissioning and intensity ramp-up. The period from week 21 to week 25 is proton physics. The blue block in week 25 represents the machine development (MD) period, while the one-week technical stop is indicated in green. (Image: CERN)

Proton collisions will be paused in the last week of June for a technical stop, followed by a special physics run in early July. For the first time ever, the LHC will inject, accelerate and collide oxygen ions. After a four-day set-up period, four days of oxygen–oxygen collisions will provide data for physics.

Once the oxygen run is complete, a two-day calibration run ([Van der Meer scans](#)) will mark the return of proton physics, paving the way for a productive summer and early autumn of data collection.

The proton run is scheduled to end on 3 November and the lead-ion run to start on 15 November, for a period of 21 days, followed by two days of machine development (MD). At 6.00 a.m. on 8 December, the final beams of 2025 will be dumped, marking the end of operations and the start of the next YETS.



The last two quarters of the approved 2025 LHC schedule. The oxygen run and Van der Meer scans in weeks 27 and 28 will be followed by a long period of proton physics from week 29 to week 45. The blue blocks indicate machine development (MD). The lead-ion set-up will start in week 45 and finish in week 46, leading into the lead-ion physics run, which will end on 8 December. (Image: CERN)

Of course, for the LHC to be ready to start the recommissioning with beam on 5 April, the injector chain must be recommissioned well in advance. In fact, the Linac4 source is scheduled to start up next week, on 29 January, in order to deliver beam to Linac4 on 19 February. A week later, on 27 February, the Linac4 beam will be injected into the PS Booster. Shortly after, on 5 March, the PS Booster will transfer its beam to the PS for the latter’s beam recommissioning. The SPS will receive the beam on 14 March and then prepare the LHC beam in time for delivery to the LHC before 5 April.

Meanwhile, each machine in the injector chain will also prepare beams for the fixed-target

experiment facilities. These include ISOLDE, the PS East Area, n_TOF and the SPS North Area. Beams will also be prepared for the Antiproton Decelerator (AD) target, which is crucial for producing the antiprotons needed by the AD and ELENA decelerators. The ELENA machine is then expected to deliver low-energy antiprotons for physics to its experiments on 28 May. The AWAKE and HiRadMat beams will also be prepared for first delivery on 21 April and 5 May, respectively.

As for ion operations, Linac3 and LEIR will start up early in April 2025, so as to be ready to deliver oxygen-ion beams to the PS on 26 June. The PS will transfer these beams to the SPS by 10 June in preparation for the four-day oxygen physics run in the LHC and the nine-day oxygen physics run in the SPS North Area, starting on 9 July.

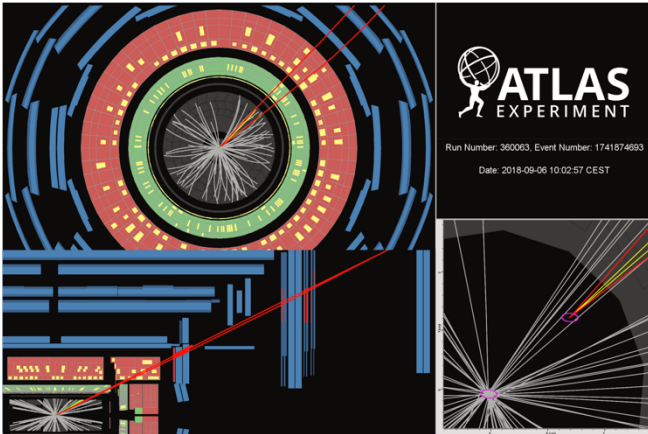
After the conclusion of the oxygen run, the injector chain will be converted to deliver lead ions. These will be supplied to the LHC, the SPS North Area and the PS East Area, before the injectors also shut down at 6.00 a.m. on 8 December.

But before looking ahead to the end of the run, let’s focus on what promises to be an exciting and successful year and, of course, stay tuned for more news on the recommissioning of our accelerator complex...

Rende Steerenberg

The measurement of a lifetime

New ATLAS measurement of the lifetime of the electrically neutral beauty meson is the most precise to date



Display of a candidate neutral beauty meson decaying into an excited neutral kaon and a J/ψ meson. The J/ψ meson decays into a pair of muons (red lines) and the excited neutral

kaon decays into a charged pion and a charged kaon (yellow lines). (Image: ATLAS/CERN)

The [ATLAS](#) collaboration at the Large Hadron Collider ([LHC](#)) has released a [new high-precision measurement](#) of the lifetime of the electrically neutral beauty (B^0) meson – a hadron composed of a bottom antiquark and a down quark. Beauty (B) mesons are made up of two quarks, one of which is a bottom quark. Over the past decades, by studying B mesons, physicists have been able to examine rare and precisely predicted phenomena to gain insights into interactions mediated by the weak force and into the dynamics of heavy-quark

bound states. The precise measurement of the B^0 meson lifetime – the average time it exists before decaying into other particles – is of critical importance in this context.

The new ATLAS study of the B^0 meson looked for the particle’s decay into an excited neutral kaon (K^{*0}) and a J/ψ meson. The J/ψ meson subsequently decays into a pair of muons while the K^{*0} meson is studied through its decay into a charged pion and a charged kaon. The analysis is based on proton–proton collision data collected by the ATLAS detector during Run 2 of the LHC (2015–2018), amounting to an impressive data set of 140 inverse femtobarns (1 inverse femtobarn corresponds to approximately 100 trillion proton–proton collisions).

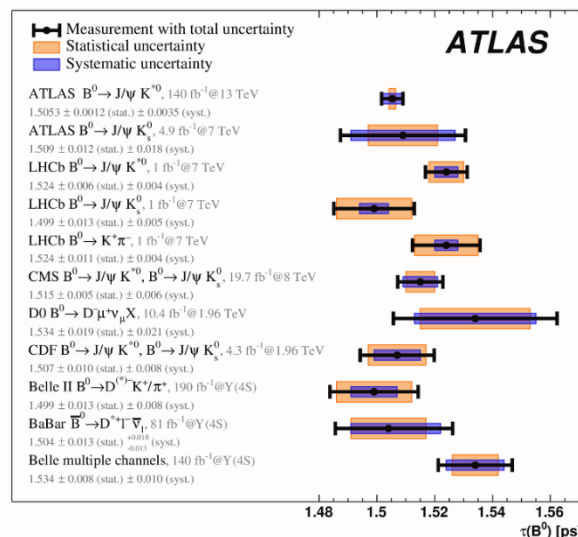
The ATLAS researchers measured the B^0 meson lifetime to be 1.5053 picoseconds (1 picosecond (ps) is a trillionth (10^{-12}) of a second), with a statistical uncertainty of 0.0012 ps and a systematic uncertainty of 0.0035 ps. This result is the most precise to date and significantly improves upon previous measurements, including a previous ATLAS result (see figure below).

To achieve such precision, many experimental challenges needed to be overcome, including minimising systematic uncertainties, performing precise modelling and improving detector alignment.

In addition to measuring the lifetime of the B^0 meson, the ATLAS team calculated its decay “width”. Width is a fundamental parameter of any unstable particle with a finite lifetime. The shorter the lifetime, the broader the decay width – a direct consequence of Heisenberg’s uncertainty relation in quantum mechanics. The decay width of the

B^0 meson was measured to be 0.664 inverse picoseconds (ps^{-1}), with a total uncertainty of 0.004 ps^{-1} .

The researchers then compared this result to a [previous measurement](#) of the decay width of the B_s^0 meson (composed of a bottom quark and a strange quark). They found that the ratio of the decay widths was consistent with unity, reflecting the close values of the measured widths. These results are in line with heavy-quark model predictions and can be used to further tune these predictions.



A comparison of the new ATLAS result for the B^0 lifetime with the previous ATLAS result, as well as results from the LHCb, CMS, D0, CDF, Belle II, BaBar and Belle experiments. (Image: ATLAS/CERN)

The new ATLAS precision measurements enhance the understanding of weak-force-mediated decays in the Standard Model and provide valuable data for future theoretical developments.

ATLAS collaboration

LHCb sheds light on two pieces of the matter–antimatter puzzle

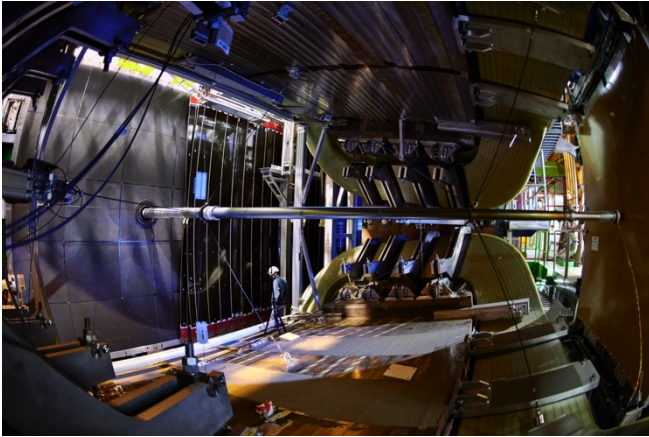
The collaboration has found evidence of CP violation in decays of baryons and in decays of beauty hadrons into charmonium particles

In the Big Bang, matter and [antimatter](#) should have been created in equal amounts. But fast forward 13.8 billion years to the present day and the Universe is made almost entirely of matter, so something must have happened to create this imbalance.

The [Standard Model](#) of particle physics predicts an asymmetry between matter and antimatter

known as charge–parity (CP) violation. But the size of this asymmetry in the Standard Model is not large enough to account for the imbalance and the asymmetry has so far been observed only in certain decays of particles called mesons, which are made of a quark and an antiquark. It remains to be seen in other meson decays and in decays of

other types of particles, such as three-quark particles called baryons.



The LHCb detector seen in 2018 during its opening. (Image: CERN)

In two new articles, the [LHCb](#) collaboration at the [Large Hadron Collider](#) (LHC) reports seeing evidence of CP violation in [decays of baryons](#) and in [decays of beauty hadrons into charmonium particles](#), shedding light on these two pieces of the matter–antimatter puzzle.

Experiments, including LHCb, have previously conducted searches for CP violation in baryons, by looking for differences in the way matter and antimatter baryons decay into other particles. But these searches have essentially come up empty handed so far. Although one LHCb [study](#) has delivered evidence of the process in a particular decay of the bottom lambda baryon, that evidence has not increased in a subsequent [study](#) that analysed a larger sample of such decays.

In its [first](#) new study, the LHCb team sifted through proton–proton collision data taken during the first

and second runs of the LHC to search for different decay modes of the bottom lambda baryon, including its decay into a lambda baryon and two kaons. It then looked for CP violation in each decay mode by basically counting the number of decays of the bottom lambda baryon and of its antimatter partner and taking the difference between the two. For the decay into a lambda baryon and two kaons, this difference showed evidence of CP violation with a significance of 3.2 [standard deviations](#).

In its [second](#) new study, the LHCb team turned its attention to the decay of the electrically charged beauty meson into a J/psi and a charged pion. The J/psi is a charmonium particle – a meson consisting of a charm quark and a charm antiquark. Performing a similar analysis to that of the bottom lambda baryon study, and also using data from the first and second runs of the LHC, the LHCb researchers found evidence of CP violation in this decay mode of the charged beauty meson, again with a significance of 3.2 standard deviations. The finding represents evidence of CP violation in decays of beauty hadrons to charmonium particles.

The two studies mark significant steps towards establishing whether or not CP violation exists in these types of decays. Data from the third run of the LHC and from the collider’s planned upgrade, the [High-Luminosity LHC](#), are set to shed further light on these and other pieces of the matter–antimatter puzzle.

Ana Lopes

Work Well Feel Well: Emotions: forces for transformation

Part eight of the [Work Well Feel Well series](#) looks at how to acknowledge and manage our emotions

How can we manage frustration, annoyance or aggression at work?

It’s natural for us to experience emotions. Emotions are universal, they are intense and they are beyond the scope of our reason. Being able to live healthily with our emotions is a sign of physical vitality and flexibility.

Work can impose a demanding framework on our emotional life: we need to know how to assert

ourselves, be motivated and, sometimes, be discreet, all the while rolling with the punches. We need to shoulder the responsibility of managing our emotions so that we can work efficiently. Norms in the workplace differ from those in our private lives. We may give ourselves permission for outbursts of anger towards those closest to us from time to time, but there is a danger that anger

could be transformed into frustration or passive aggression at work.

If suppressed and unacknowledged, emotions accumulate and manifest themselves through breaches of trust, inappropriate irony, resistance to change, lack of sensitivity in relationships, or a negative and demeaning attitude towards others – with disastrous consequences for efficient cooperation and team spirit. Containment is an intelligent, fast and economical way of managing emotions in the workplace. It means recognising, feeling and being aware of the emotion, delaying its expression and releasing it later on, in the right place and in a healthy manner.



What people think being emotional means: unprofessional, weak, irrational. What it actually means: you are human. (Image: Liz Fosslien)

Emotions can become powerful forces for transformation. For example, once anger has been recognised and accepted, it can be transformed into determination. We can draw support from it, and it makes us clearer and firmer. An emotion can enable us to shift focus: dissatisfaction can become a motivator of change. Some people will suffer from frustration and remain burdened by it and miserable, while others will take advantage of

the frustration to move onto something else, by extracting the positive energy needed for transformation.

“There are no positive or negative emotions, just emotions we feel when our needs are met and emotions we feel when our needs are not met.”

-Marshall Rosenberg

Take action

As part of the “Efficiency and caring at work” campaign, the [Work Well Feel Well website](#) now offers useful resources that can be downloaded, including an [exercise](#) to assess how you manage your emotions at work.

Other suggestions include:

- **Managing self:** one of the behavioural competencies of the [CERN Competency Model](#) is “[managing self](#)”, which includes understanding our own emotions and how they impact behaviour in effective and ineffective ways. This theme is also explored in the recent Ombud text “[Practice responding instead of reacting](#)”.
- **Feelings and needs inventory:** The Center for Nonviolent Communication offers a free downloadable [inventory](#) to help us to understand our emotions.
- **Emotional intelligence:** watch the recording of the micro-talk entitled “[Cultivating Emotional Intelligence \(EI\) in the Era of Artificial Intelligence \(AI\)](#)” to hear about the importance of emotional intelligence for human connection, creativity and innovation.
- **Medical service support:** consult the medical service’s dedicated [mental health support webpage](#) for information, resources and guidance.
- **Training courses:** the [CERN training catalogue](#) includes a course on how to [develop emotional agility](#) to help to build trust in emotionally charged situations.

HR department

Computer Security: Are you a team player?

The field of research is a team sport. Long gone are the days when one individual could make the change, the discovery of the century, and win the big prize. Today, we stand on the shoulders of giants, of giant teams, of collaborations, and only jointly can we reach our (research) goals. But when push comes to shove, are you willing to take one for the team?

Actually, this is a serious question. Recently, we detected that a colleague's home computer had been compromised by "Infostealer", which was exfiltrating credentials from Google Chrome. The PC was thoroughly infected and all the passwords ever typed on it were exposed and stolen. Following that detection, we sent a direct message and asked our colleague to not only reinstall the PC, but also change all the passwords ever used on it. But the latter request was ignored. And an important password linked to a test instance of a critical CERN computing service was among the passwords compromised. The attackers were quick to realise that blunder and started (trying to) take over that test instance. Fortunately, the production instance was spared...

Thus, a simple incident at home was carried over into CERN, putting CERN at risk. Despite the fact that we explicitly asked the colleague to make sure that:

○ I have reinstalled every device involved, e.g. those compromised and those where any compromised password has been actively used. AFTERWARDS, I have [changed my CERN password](#) and any other password actively used on those devices. Finally, I have checked all attached storage systems, e.g. CERNbox, EOS, Network Attached Storages (NAS) for any suspicious activities or changes.

Admittedly, this is a tedious and cumbersome task. Who remembers all the passwords ever used on their device apart from those that are most regularly used? (Check out your Chrome password store!) And who is prepared to go through that list and reset them all? Who actually wants to reinstall that infected device, given that this might imply loss of precious data (if no back-up exists) and some significant work to reinstall its operating system and applications? Who is willing to take that punch for the team?

Evidently, the initial problem was homegrown. Some inattentiveness, some forgotten "STOP – THINK – DON'T CLICK" while browsing or opening emails, some reluctance to keep the PC up-to-date, or some kids or partners using the same PC and committing their own blunder... Bad luck. But the consequences go beyond "home". They affect CERN. And the colleague concerned should assume their responsibilities: reinstall, change passwords, take the punch.

But instead, avoidance. Head in the sand. Letting fate take its course. A compromised test system. A near miss for its production system. Putting CERN at risk... What would you do? Are you a team player?

Computer Security Office

Official News

Members of the personnel shall be deemed to have taken note of official news published for the CERN community. Reproduction of all or part of this information by persons or institutions external to the Organization requires the prior approval of the CERN Management. The full list of official news is available here: <https://cern.ch/official-news>.

Remuneration payment dates in 2025

In 2025, salaries of staff members, stipend of fellows and graduates, contractual subsistence allowances of associated members of personnel will be paid on the following dates:

- Friday 24 January

- Tuesday 25 February
- Tuesday 25 March
- Friday 25 April
- Monday 26 May
- Wednesday 25 June

- Friday 25 July
- Monday 25 August
- Thursday 25 September
- Friday 24 October

- Tuesday 25 November
- Thursday 18 December

FAP department

Annual adjustments to financial benefits with effect from 1 January 2025

In accordance with recommendations made by the Finance Committee and decisions taken by Council in December 2024, the following financial benefits will be adjusted, with effect from 1 January 2025:

- A 1.21% increase to the scale of basic salaries paid to Staff Members and the scale of stipends paid to Fellows and Graduates (Annexes R A 5 and R A 6 of the Staff Regulations).
- A 0.6% increase to the subsistence allowances (Annex R A 7 of the Staff Regulations), family, child and infant allowances (Annex R A 3 of the Staff Regulations) and to payment ceilings of education fees (Annex R A 4 of the Staff Regulations).

The index of 2.24% for home leave has already been applied as of June 2024.

The new ceiling of education fees will be applied to the entire school year 2024-2025.

Other related adjustments have been implemented wherever applicable to associated members of the personnel ([2025 subsistence rates](#)).

The amended text of the Staff Regulations will be available shortly on the Web at: [CERN Staff Rules and Regulations](#).

HR department

CERN Health Insurance Scheme (CHIS) – Monthly contributions as of 1 January 2025

The CHIS contribution rates are unchanged for 2025, but the contributions themselves will evolve due to the change in the relevant Reference Salary (see [Chapter XII of the CHIS Rules](#)). Thus, as of 1 January 2025, the lump-sum monthly contributions based on Reference Salary II will be as follows:

1. Lump-sum contributions for voluntary members

The monthly contribution for voluntary members (e.g. users and associates) with the normal health insurance will be 1 287 CHF per month, whilst for

those with the reduced health insurance it will be 644 CHF.

2. Lump-sum contributions for post-compulsory members other than CERN pensioners

For post-compulsory members other than CERN pensioners, the monthly contribution will be 1 375 CHF in the case of former staff members awaiting a deferred pension and former spouses continuing their affiliation, whilst in the case of formerly dependent children continuing theirs it will be 550 CHF.

HR department

Staff Rules and Regulations - 11th edition

Modification No. 25

In accordance with recommendations made by the Finance Committee and decisions taken by Council in December 2024 (CERN/FC/6866-CERN/3861), please find below the pages of the Staff Rules and Regulations which have been updated further to the modifications coming into force on 1 January 2025:

- Annex R A 3 of the Staff Regulations, (Family allowances), *modification of page 68*
- Annex R A 4 of the Staff Regulations, (Education fees), *modification of page 69*
- Annex R A 5 of the Staff Regulations, (Monthly basic salaries of staff members), *modification of page 71*

- Annex R A 6 of the Staff Regulations, (Stipends of graduates and fellows), *modification of page 72*
- Annex R A 7 of the Staff Regulations, (Subsistence allowances of associated members of the personnel), *modification of page 73*

The complete updated electronic version of the Staff Rules and Regulations is accessible via [CDS](#).

HR department

Composition of the Joint Advisory Appeals Board (JAAB/CPCR)

	<u>Appointed by the Director-General</u>	<u>Appointed by the Staff Association</u>
Members	Valeria PEREZ REALE / TE	François DUVAL / EP
1st deputies	Joël CLOSIER / EP	Nicolas SALOMON / PF
2nd deputies	Raymond VENESS / SY	Silvia SCHUH- ERHARD / BE

Valeria Perez Reale and François Duval have drawn up the following list of staff members from among whom the Chairperson of the Board may be chosen when required:

Jan BORBURGH / SY	Pedro MARTEL / EN
François BRIARD / IR	Jesper NIELSEN / BE
Isabel FERNANDEZ GONZALEZ/ SCE	Chiara PASQUINO / SY
Silvia GRAU / EN	Laurent ROY / EP
Nils HOIMYR / IT	Jens VIGEN / RCS

Mediators [see Administrative Circular N° 6 (Rev. 1) entitled "Review procedure"] will also be selected from this list of ten staff members.

Revised computer security rules

At its [second meeting](#), the [Computer Security Board](#) has formally approved several Subsidiary Rules to the [Use of CERN computing facilities \(Operational Circular No. 5, OC5\)](#) related to

[Identities, Authentication & Authorization \(IAA\)](#). The approved rules (IAA2/3/4/6/7) are now documented on the website of CERN's Computing Rules (<https://cern.ch/ComputingRules>).

At its [third meeting](#), the Computer Security Board has formally approved several Subsidiary Rules to OC5 related to [Endpoints \(EPT\)](#) and [Software Restrictions \(SWR\)](#). The approved rules (EPT1/2/4/5/6 and SWR2/3) are now documented

on the website of CERN's Computing Rules (<https://cern.ch/ComputingRules>).

For more information, read this article: [Computer Security: Security Rules revised](#)

Announcements

Upcoming events

Check out this curated list of upcoming events for the CERN community

17 Jan | Computing | Council Chamber | PostgreSQL | EN

[CERN PGDay 2025](#)

20–24 Jan | Computing | Main Auditorium | CERN Quantum Technology Initiative | EN

[International Conference on Quantum Technologies for High-Energy Physics \(QT4HEP\)](#)

21 Jan 13.30 | Knowledge Sharing | CERN Library | Archives, Library and Open Science events | EN

[The Voice of Science - "Particle Physics Engagement in Edinburgh: from Peter Higgs Legacy to LEGO and boardgames"](#)

23 Jan 10:00 | Knowledge Sharing | Salle Bohr (40/S2-B01) | KT Seminars | EN

[The first Cyclotron-based multi-ion project: Status, Perspectives and Challenges](#)

23 Jan 17:30 | Knowledge Sharing | Main Auditorium | Groupement des Français du CERN | FR

[Airbus Industrie et le CERN](#)

23–24 Jan | Experiments | Council Chamber | AIDAinnova | EN

[AIDAinnova Course on Quantum Applications](#)

23-24 Jan | Physics | Online | CERN Neutrino Platform & Physics Beyond Colliders | EN

[Workshop on Neutrinos@CERN](#)

28 Jan 13:30 | Knowledge Sharing | CERN Library | Archives, Library and Open Science events | EN

[Meet the authors of "Detectors in particle physics"](#)

28 Jan 14:00 | Computing | Online | Office of Data Privacy | EN/FR

[Data Protection Day 2025 "AI beyond the hype"](#)

30 Jan 16:30 | Knowledge Sharing | Main Auditorium | CERN Colloquium | EN

[Long Baseline Neutrino Experiments and the enabling role of the CERN Neutrino Platform: DUNE and ProtoDUNE](#)

30 Jan 17:00 | At CERN | Online | Mentoring@CERN | EN

[Mentoring@CERN Information Session](#)

31 Jan 14:00 | Knowledge Sharing | Online | Micro-Talks | FR

[#14 Analyse du langage non verbal : jusqu'où peut-on cacher la vérité ?](#)

5 Feb 18:00 | Knowledge Sharing | CERN Science Gateway | Public Events | EN

[2nd CERN Art and Science Summit](#)

Registration now open

10–14 Mar | Accelerators | Ferney-Voltaire | CERN Accelerator School | EN

[Basics of Accelerator Physics and Technology](#)

30 Apr – 13 May | Physics | Costa Rica | CLASHEP | EN

[2025 CERN Latin-American School of High-Energy Physics](#) *deadline for applications 10 January 2025*

15–27 Jun | Accelerators | Bulgaria | CERN Accelerator School | EN

[Pushing the Limits: Intensity Limitations in Hadron Beams](#)

[ISOTDAQ 2025 – International School of Trigger and Data Acquisition](#)

This is a curated list of events relevant to the CERN community. More events are available [here](#):

home.cern/events

Indico also shows ALL events [happening today, this week](#) and in a [calendar view](#).

If you would like your event to appear on an upcoming Bulletin events list, please contact bulletin-editors@cern.ch

Find out who won the 2024 End-of-Year Crossword

Thank you to everyone who took part in the CERN community crossword, here are the winners and the solutions

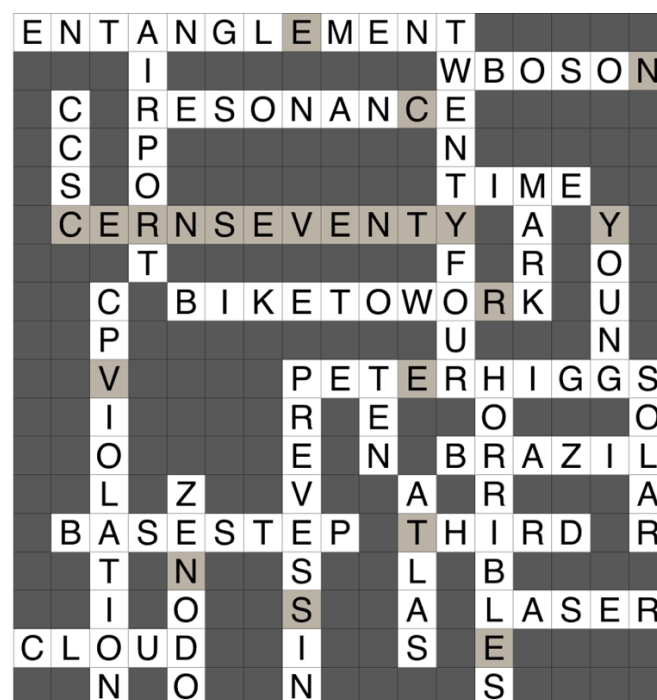


Congratulations to the three prize draw winners: Patricia Verheyden (left) won the LHC Lego set, Teng Jian Khoo (centre) won the two CAGI Balexert cinema tickets and Katharine Leney (right) won the CERN70 merchandise (Image: CERN)

Thank you to everyone who participated in the 2024 end-of-year crossword for the CERN Community. We hope that you had fun and enjoyed looking back on the past year's achievements with us.

Congratulations to Patricia Verheyden (IR-ECO), Teng Jian Khoo (EP-ADT) and Katharine Leney (EP-UAT), who all had correct entries and were selected in the prize draw.

The full solutions can be found below :



Internal Communication

Sign up for the 2025 Mentoring@CERN programme

Following the success of last year's pilot round, the LHC Early Career Mentoring Committee and CERN's [Women in Technology](#) network are thrilled to announce the continuation of the Mentoring@CERN programme in 2025.

This unique initiative aims to foster professional development and support within the CERN community by connecting mentors and mentees across all disciplines and career levels.

A look back at 2024: A successful pilot round

In its inaugural pilot phase, Mentoring@CERN brought together 91 mentor–mentee pairs, spanning departments and disciplines across CERN. The programme enabled participants to form meaningful connections, share expertise and achieve their professional goals. Participants praised the programme for its personalised matching, its flexibility and the meaningful connections it fostered. Feedback from surveys showed that over 87% of responders found the

experience to be beneficial for both professional and personal growth.

Who can participate?

If you are associated with CERN through a contract – whether as a staff member, fellow, student, user, PART or external contractor – you are eligible to take part.

The programme is open to everyone regardless of age, career level, gender or department. Mentors and mentees are carefully paired on a case-by-case basis to ensure personalised matches that meet individual goals and preferences.

Timeline for 2025

- Information session: 30 January 2025
- Learn about the programme, ask questions and discover how mentorship can enhance your career journey
<https://indico.cern.ch/event/1497017/>
- Registration period: 1–28 February 2025
- Submit your application to become a mentor, mentee or both
<https://indico.cern.ch/event/1479276/>
- Matching process: March 2025
- We will carefully review applications and announce matches by the end of March.

- Kick-off meeting: Beginning of April 2025
- This session will share mentoring best practices, templates and guidance to help you start your mentorship journey.
- Mentoring round: April–December 2025
- Once matched, mentors and mentees are free to set their own schedules and meeting frequencies.

Behind the scenes

Mentoring@CERN is a collaborative effort that brings together the expertise of two established programmes:

- The **WIT Mentoring programme**, launched in 2018, which emphasised pairing mentors with female mentees.
- The **LHC Early Career Mentoring programme**, initiated in 2020, which focused on connecting professionals across LHC experiments.

By merging these initiatives, we aim to make mentorship opportunities accessible to everyone in the CERN community.

For any questions or further information, attend our [information session](#) on 30 January 2025 or contact us at mentoring.cern@cern.ch.

Using portable heaters at CERN

As temperatures drop, portable heaters can provide extra warmth in the workplace. Portable electric heaters were removed from the CERN Stores at the end of 2022 as an energy-saving measure. However, as some buildings are less well insulated than others and people's tolerance of lower temperatures varies, many of us still use portable heaters, procured before 2022 or from outside CERN, to keep warm. Their use comes with significant safety risks, particularly relating to fire.

Portable heaters should be used as little as possible. To ensure their safe use, we ask all users to follow the guidelines below:

- Do not use open-fire, solid fuel or liquid fuel heaters.
- Make sure that there are no major fire risks in the same area (e.g. storage/manipulation of easily combustible solids or flammable liquids). Position the heater **at least one metre** away

from flammable objects such as curtains, clothing and furniture.

- Never cover a heater.
- Follow manufacturer instructions for proper usage.
- Avoid leaving heaters on unsupervised and make sure that they are switched off outside of working hours. Use a timer if necessary.
- Electrical safety: overloaded cords and sockets can overheat, posing fire risks. Do not plug more than one heater into one socket and ensure that the heater's wattage **does not exceed 1800 W for a 10 A socket**. If an extension cord is necessary, it must be fully unwound.
- Emergency preparedness: know the location of the nearest fire extinguisher (within 40 metres) and familiarise yourself with the general

evacuation procedures applicable to your workstation.

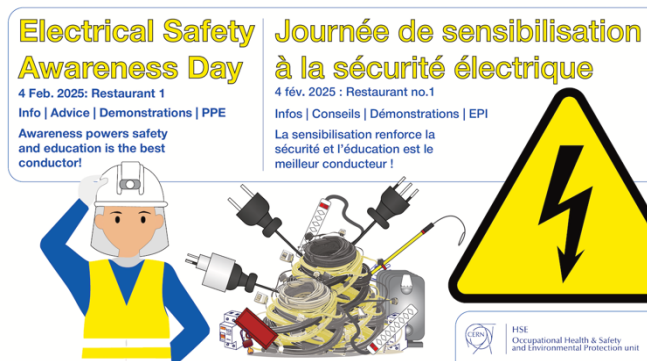
By following these precautions, you help to keep your environment safe. For further guidance,

contact your Territorial Safety Officer (TSO). Stay warm and safe this winter!

HSE unit

Electrical Safety Awareness Day

Join us on 4 February in Restaurant No. 1 to learn more about electrical risks and how to keep yourself and others safe



Join us on 4 February in Restaurant No. 1 to learn more about electrical risks and how to keep yourself and others safe (Image: CERN)

Electricity powers our world, but what's your level of confidence with it? Are you switched on to electrical safety best practices, or could your knowledge use a spark to ensure you and others stay safe?

We all interact with electricity every day, but we may not be fully aware of the risks it poses. Whether your role involves something as simple as using a Swiss adaptor for your laptop or as

advanced as working hands-on with CERN's intricate electrical installations, the electrical risks involved could have serious consequences. Reflecting on these risks and learning how to manage them effectively is time well spent.

Join us in **Restaurant No. 1 by the Grab & Go on 4 February from 10 a.m. to 3 p.m.** for live advice, demonstrations and practical information from CERN's electrical safety experts. Whether you're curious about the risks of travel adaptors and extension leads or need guidance on selecting your next arc-flash visor or personal protective equipment (PPE), this is your opportunity to learn, ask questions and enhance your workplace safety. Keep yourself and your equipment safe at home, at work or wherever you're next plugging in by taking a few minutes to join us and benefit from the advice of the CERN experts.

Awareness powers safety, and education is the best conductor.

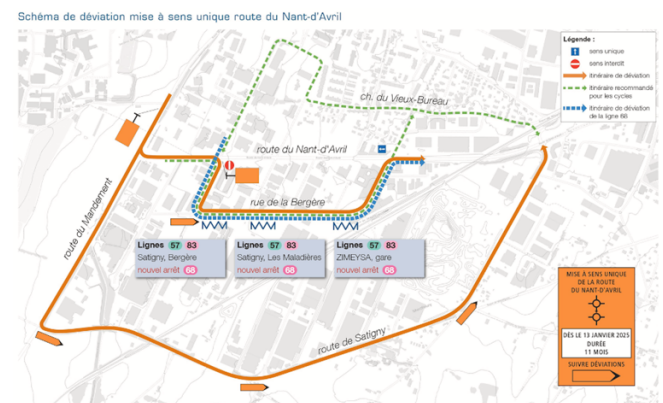
HSE unit

Works on the Route du Nant-d'Avril from 13 January 2025

Roadworks on the Route du Nant-d'Avril will be carried out by the canton of Geneva from 13 January 2025, for approximately 11 months.

Traffic on Route du Nant-d'Avril to operate in one direction between the roundabouts of Rue de la Bergère and Rue de Veyrot, in the direction of Route du Mandement. A diversion for vehicles will be put in place, as shown on the map below:

The canton of Geneva is carrying out these works to upgrade the Route du Nant-d'Avril for the Geneva - Vernier - ZIMEYSAVER fast bus service (BHNS-GVZ) by 2027.



For more information, please consult the pages
Avis de travaux : Info mobilité - BHNS-GVZ - Route
du Nant-d'Avril IV (links
<https://www.ge.ch/node/38134>;
<https://www.ge.ch/blog/bhns-gvz>).

Host State Relations service
Tel.: 75152
Relations.secretariat@cern.ch

Sign up for CERN's Environmentally Responsible Procurement e-learning course

We can all contribute to CERN's efforts to minimise the environmental impact of procurement

As part of its commitment to [environmentally responsible research](#), CERN published its [Environmentally Responsible Procurement Policy](#) in 2023, with accompanying measures to positively influence procurement-related emissions, which represent approximately [one third](#) of the Organization's total greenhouse gas emissions.

In this context, the IPT department has developed a short, 20-minute e-learning course, which provides an overview of the Organization's strategy with respect to the environment and of CERN's current carbon footprint. It introduces the diverse concepts related to environmentally responsible procurement and describes the best practices to be followed in line with CERN's associated policy.

The e-learning course aims to equip everyone involved in the procurement of supplies or services with the basic knowledge needed to make more environmentally conscious purchasing decisions and thus contribute to the implementation of the policy. Course participants will learn how environmentally responsible procurement decisions will help reduce CERN's environmental impact and consumption of resources, thereby supporting the Organization's environmental objectives and ultimately its resilience.

Sign up to the course now on the CERN learning hub: <https://cern.ch/ceerp-elearning> and make your contribution to environmentally responsible procurement at CERN.

IPT department